

Cost of Living (US)- Data Visualization Analysis

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Introduction

- **Cost of Living:** The amount of money required to maintain a specific standard of living, including essentials like housing, food, healthcare, and taxes
- Cost of living is complex and multidimensional

Quantitative:	Categorical:
Housing Cost	State
Food Cost	isMetro
Transportation Cost	Area Name
Healthcare Cost	County
Other Necessities Cost	Family Member Count
Childcare Cost	
Taxes	
Total Cost	
Median Family Income	



Problem/Objectives

Problem statement:

Cost of living varies by state, but the patterns/trends are not always obvious without complex analyzation

Objectives:

- Identify trends/patterns across states
- Identify correlations between variables
- Identify Outliers
- Analyze geographic patterns



Data Visualization Overview

- Data Cleaning: Dropped null values (10 rows of data)
- Tools used: Python and Plotly
- Visualization Approach:
 - Interactive Tables
 - Bar Charts
 - Choropleth Maps
 - Interactive Scatter Plots
 - Correlation Matrix
 - Regression Line
 - Bubble Plots
 - 3D Plots
- These will help identify trends and patterns in the dataset



Interactive Table

- Currently sorted alphabetically by state
- Can click on arrow next to each variable to sort by highest/lowest cost

state	housing_cost	food_cost	transportation_cost	healthcare_cost	other_necessities_cost	childcare_cost	taxes	total_cost	median_family_income
AK	15520.43	8694.78	12361.72	15641.82	8774.13	15194.39	9047.29	85234.56	81242.83
AL	9181.39	8480.71	13141.57	14236.29	6399.68	9552.57	7975.35	68967.57	58099.87
AR	8886.49	7788.49	13131.93	12038.51	6042.00	7732.10	5872.37	61491.89	55025.57
AZ	12793.91	8146.29	13479.58	14532.48	7587.46	11314.89	8131.32	75985.94	59661.88
CA	20232.49	9673.95	14740.17	14221.49	10836.28	10761.33	12713.44	93179.16	82922.17
CO	14133.44	9168.69	14616.03	10427.42	8443.28	11290.98	9515.95	77595.78	75158.81
CT	16953.56	9703.83	13606.71	13085.25	9659.02	13624.39	13380.73	90013.50	103228.23
DC	21089.39	11631.16	7961.97	14065.90	11855.94	31940.49	21744.79	120289.64	116607.39
DE	14449.79	8989.75	13320.45	14330.67	8493.07	11319.14	10602.65	81505.52	81198.66
FL	13166.64	8955.00	12972.21	13543.69	8015.54	8797.53	6071.10	71521.72	64294.18
GA	9991.27	8232.04	13208.21	12863.15	6603.02	7299.59	6701.05	64898.34	59841.34
HI	22335.14	12852.93	13949.07	11638.53	12750.02	13198.92	18172.65	104897.26	99596.99
IA	9710.73	7872.14	13895.87	12156.45	6370.97	8656.97	7425.91	66089.04	74522.31
ID	10711.26	8729.47	14391.55	12514.75	7044.14	9655.05	7459.80	70506.02	63504.20
IL	10140.77	7684.84	13523.06	15632.79	6458.92	10554.93	9029.30	73024.61	73194.43
IN	10183.02	7476.63	13385.54	12401.91	6398.79	12064.41	7251.38	69161.68	70347.77
KS	9675.84	8134.01	13828.67	14325.08	6453.21	7853.49	6962.78	67233.08	68094.71
KY	9264.44	7454.15	13444.11	12637.32	6057.80	9151.55	7284.86	65294.23	58144.30
LA	10721.37	8451.39	13140.45	14997.66	6947.05	6766.50	7257.93	68282.35	58620.87
MA	21447.66	10370.57	12986.79	11432.14	11529.00	18146.39	15979.02	101891.58	100532.21
MD	16307.28	8959.53	13674.56	9015.97	9155.16	12018.68	9870.32	79001.49	96138.13



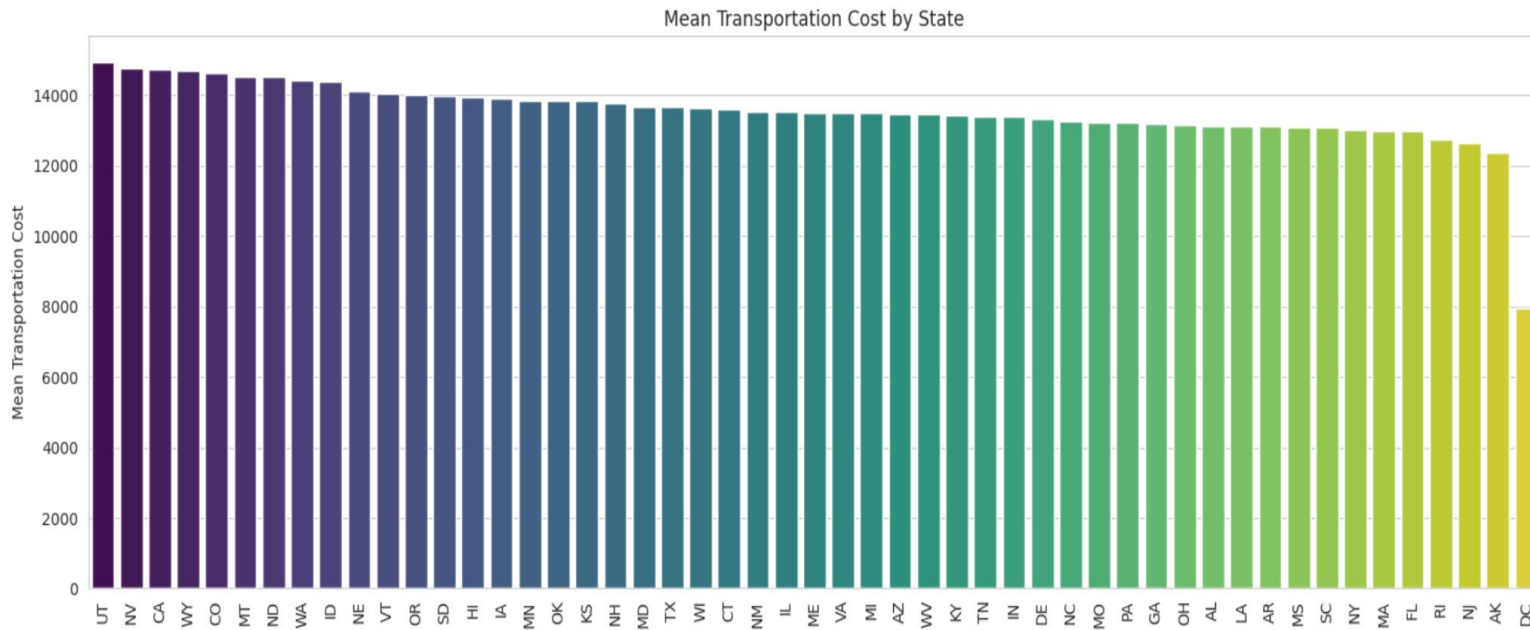
Interactive Map with Colors

- Lighter color = Higher cost
- Sorted by Median Family Income lowest to highest
- Most states with low income also have low costs
- Transportation cost is mostly green because transportation cost doesn't vary much by state

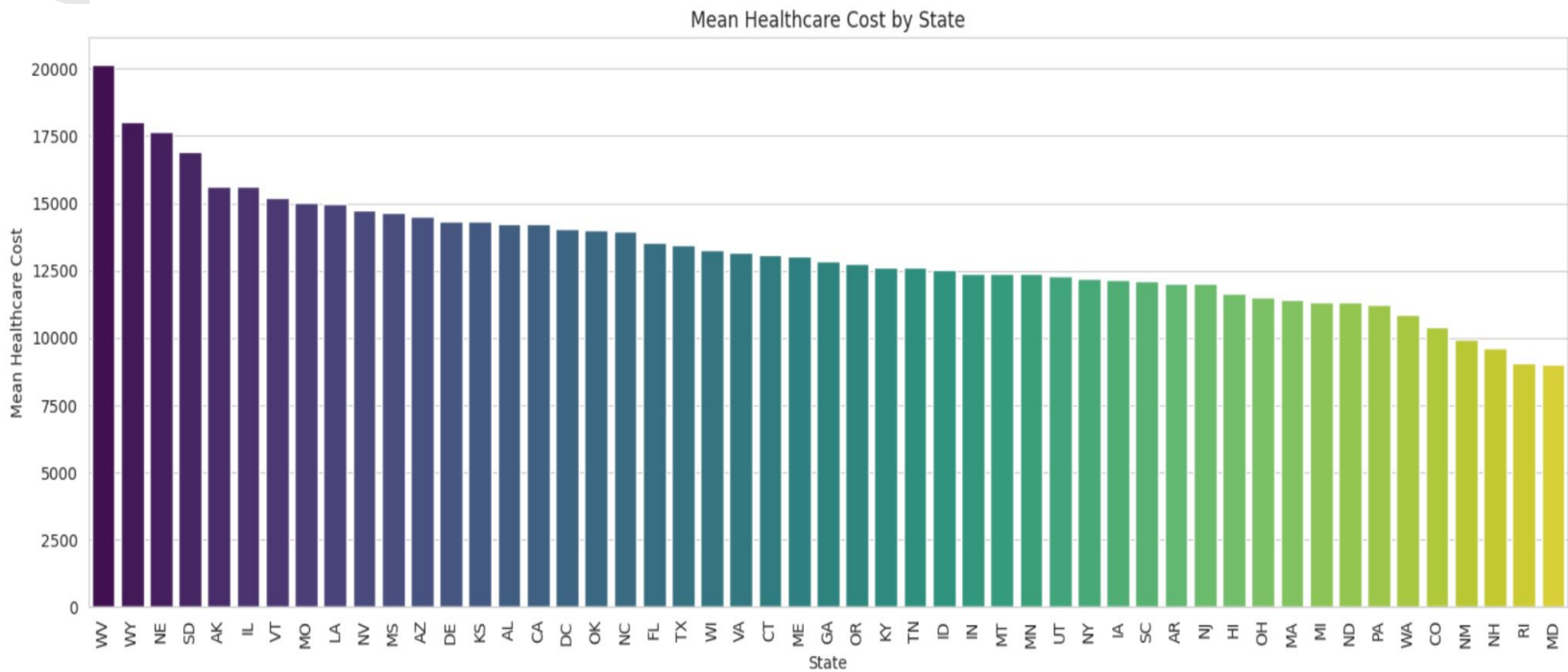
state	housing_cost	food_cost	transportation_cost	healthcare_cost	other_necessities_cost	childcare_cost	taxes	total_cost	median_family_income
MS	9350.470000	8264.920000	13080.320000	14640.990000	6382.750000	3847.000000	5794.530000	61360.990000	52281.520000
AR	8886.490000	7788.490000	13131.930000	12038.510000	6042.000000	7732.100000	5872.370000	61491.890000	55025.570000
NM	11098.560000	8407.740000	13523.320000	9936.790000	7067.900000	10937.700000	6715.550000	67687.560000	57366.030000
AL	9181.390000	8480.710000	13141.570000	14236.290000	6399.680000	9552.570000	7975.350000	68967.570000	58099.870000
KY	9264.440000	7454.150000	13444.110000	12637.320000	6057.800000	9151.550000	7284.860000	65294.230000	58144.300000
WV	9264.170000	7737.180000	13462.660000	20142.650000	6160.260000	10107.870000	10024.920000	76899.700000	58596.520000
LA	10721.370000	8451.390000	13140.450000	14997.660000	6947.050000	6766.500000	7257.930000	68282.350000	58620.870000
AZ	12793.910000	8146.290000	13479.580000	14532.480000	7587.460000	11314.890000	8131.320000	75985.940000	59661.880000
TN	9742.250000	8404.920000	13388.360000	12628.860000	6575.430000	7788.340000	4342.950000	62871.110000	59715.510000
GA	9991.270000	8232.040000	13208.210000	12863.150000	6603.020000	7299.590000	6701.050000	64898.340000	59841.340000
SC	10396.030000	7944.090000	13075.940000	12124.860000	6645.350000	6562.760000	5355.840000	62104.870000	60021.240000
MO	9243.390000	7758.860000	13244.390000	15048.010000	6160.590000	7846.440000	6822.010000	66123.690000	61569.830000
OK	9835.390000	8258.430000	13860.080000	13996.430000	6556.110000	10655.580000	7948.660000	71110.680000	61805.010000
NC	10528.100000	8101.790000	13268.160000	13978.250000	6750.340000	9440.540000	7946.410000	70013.590000	62694.320000
ID	10711.260000	8729.470000	14391.550000	12514.750000	7044.140000	9655.050000	7459.800000	70506.020000	63504.200000
FL	13166.640000	8955.000000	12972.210000	13543.690000	8015.540000	8797.530000	6071.100000	71521.720000	64294.180000
MI	10453.400000	8081.840000	13498.830000	11321.190000	6716.050000	11360.720000	7350.060000	68782.100000	66016.920000
TX	11119.060000	7673.700000	13660.940000	13461.440000	6809.360000	7362.310000	4708.590000	64795.400000	66050.290000
MT	10777.070000	8841.370000	14527.150000	12372.540000	7108.530000	11832.670000	9297.520000	74756.850000	66608.080000
KS	9675.840000	8134.010000	13828.670000	14325.080000	6453.210000	7853.490000	6962.780000	67233.080000	68094.710000



Bar Graph - Transportation Cost

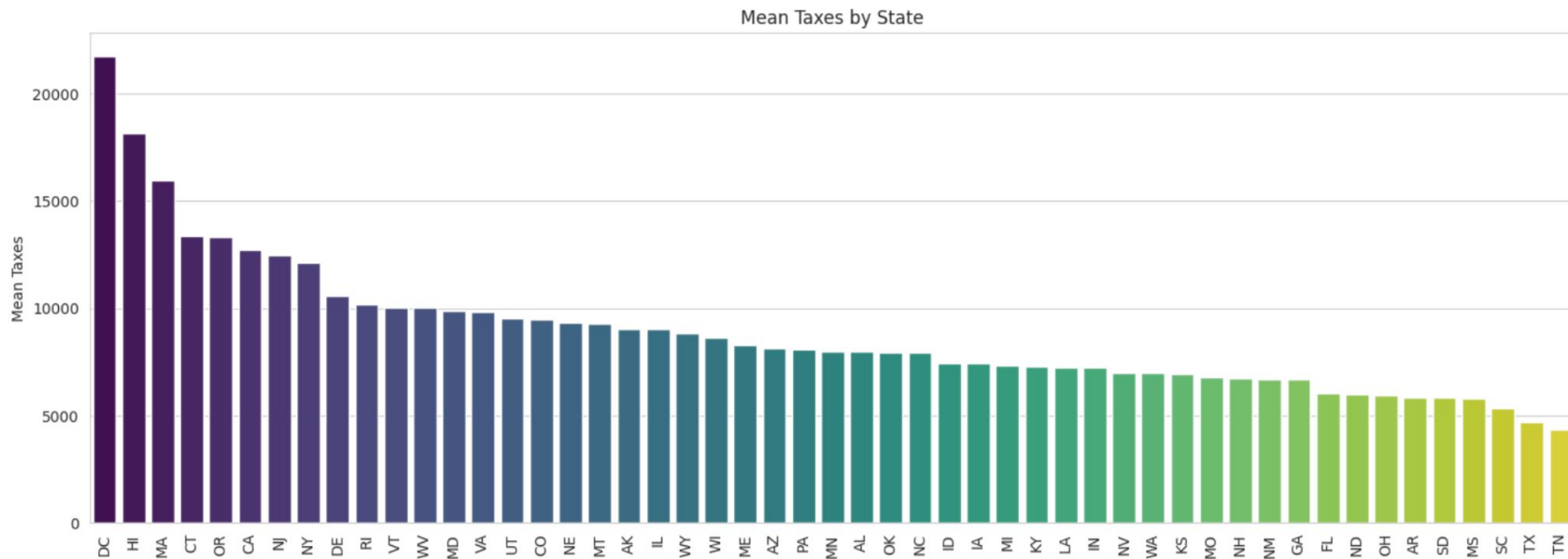


Bar Graph - Healthcare Cost



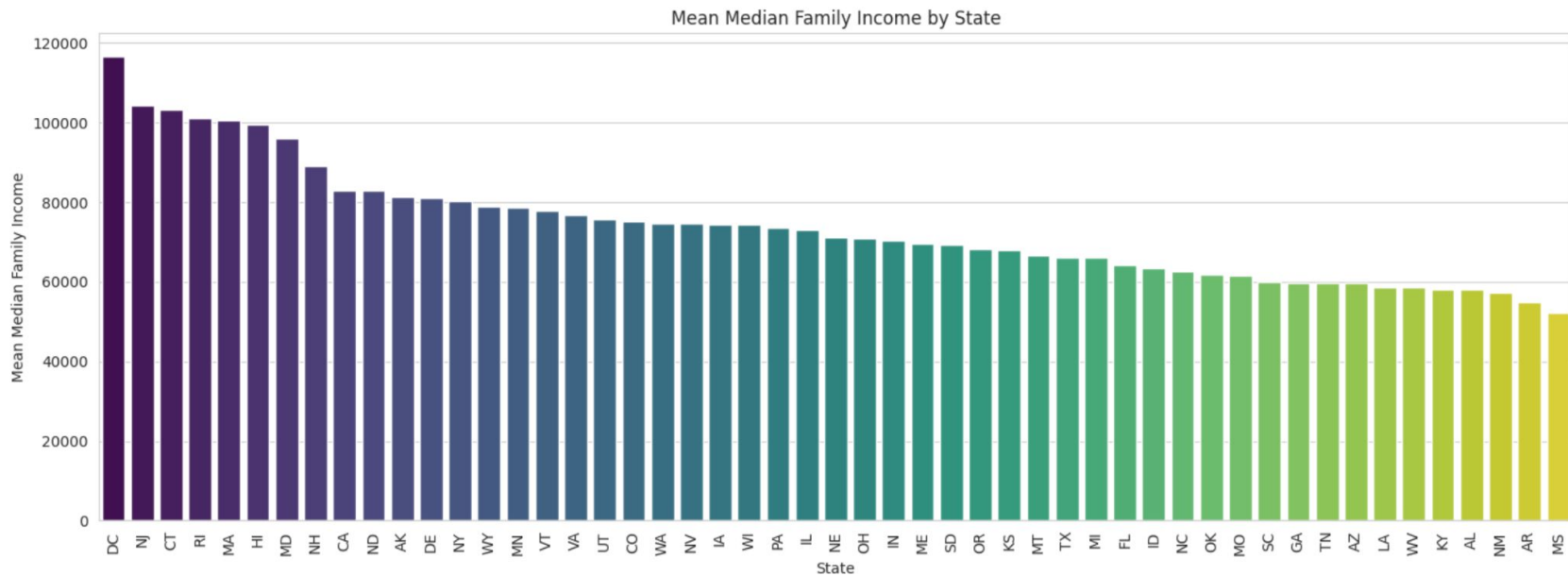


Bar Graph - Taxes



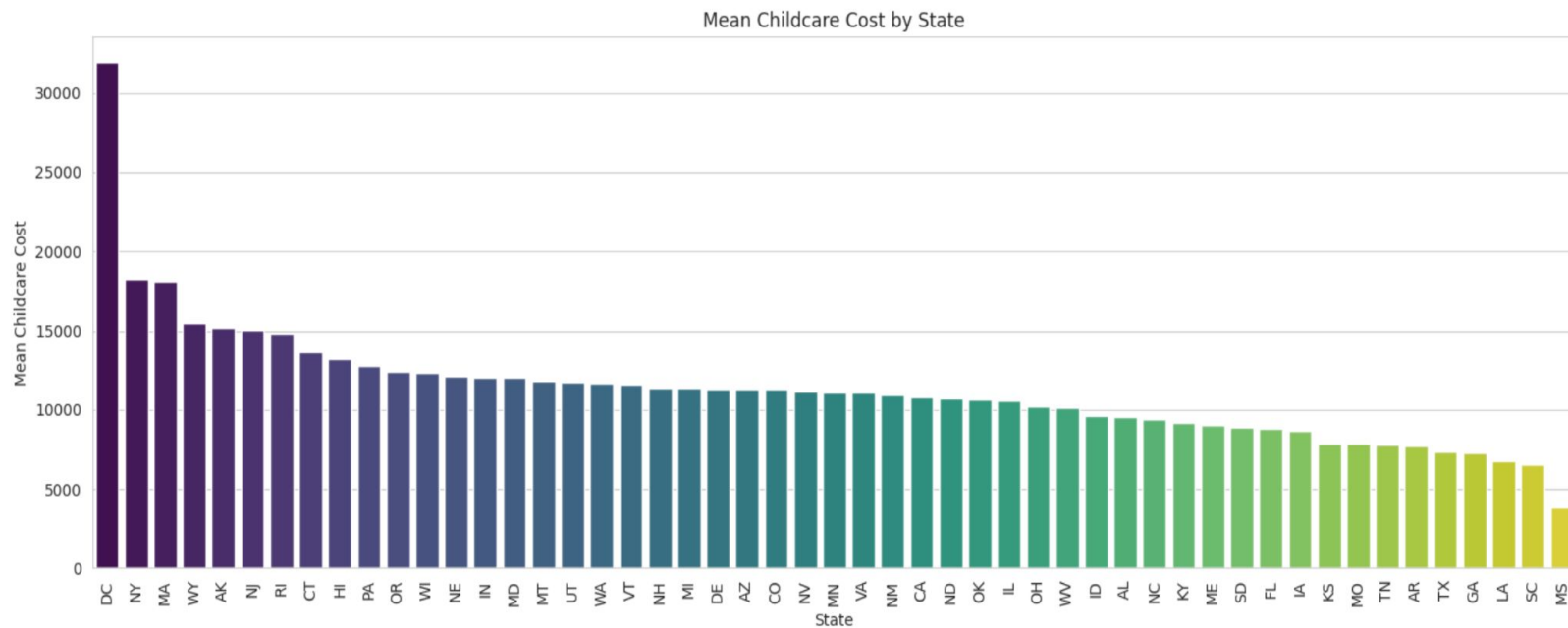


Bar Graph - Median Family Income





Bar Graph - Childcare Cost





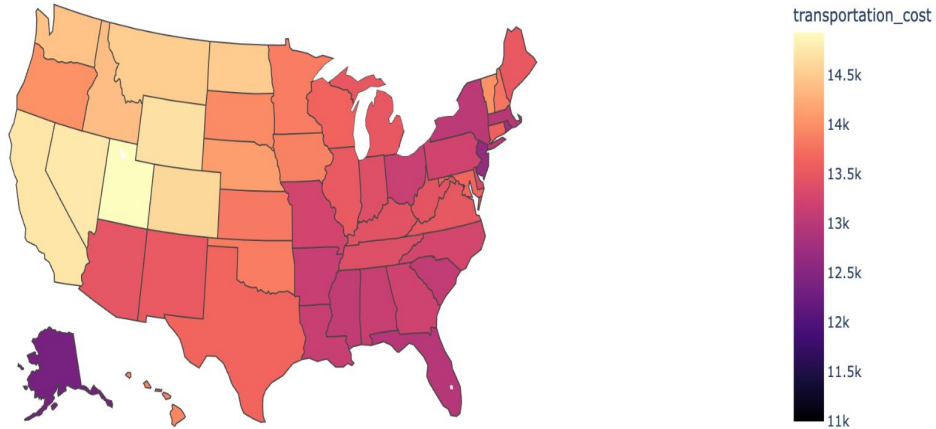
Bar Charts Summary

- Geographical trends in most charts
- Washington DC was either the highest or lowest in every graph but one
- Bar graphs are a great tool to compare variables by state



Choropleth Map - Transportation Cost

Transportation Cost by State (Choropleth Map)

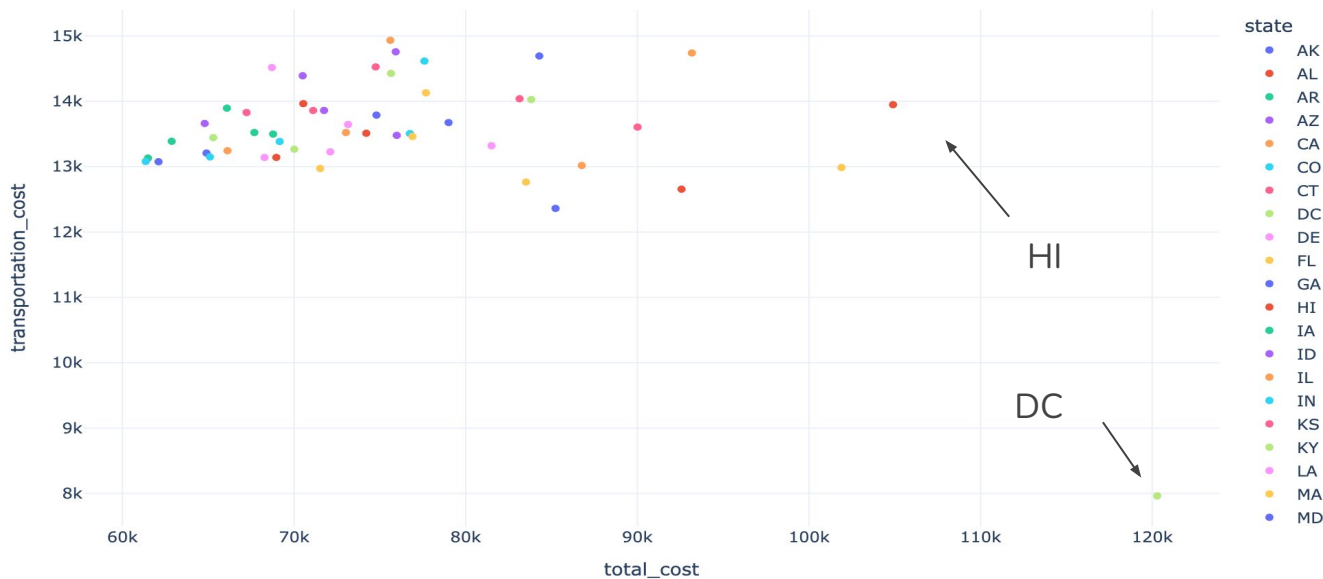


- Lighter color = Higher cost



Scatter Plot - Transportation Cost vs. Total Cost

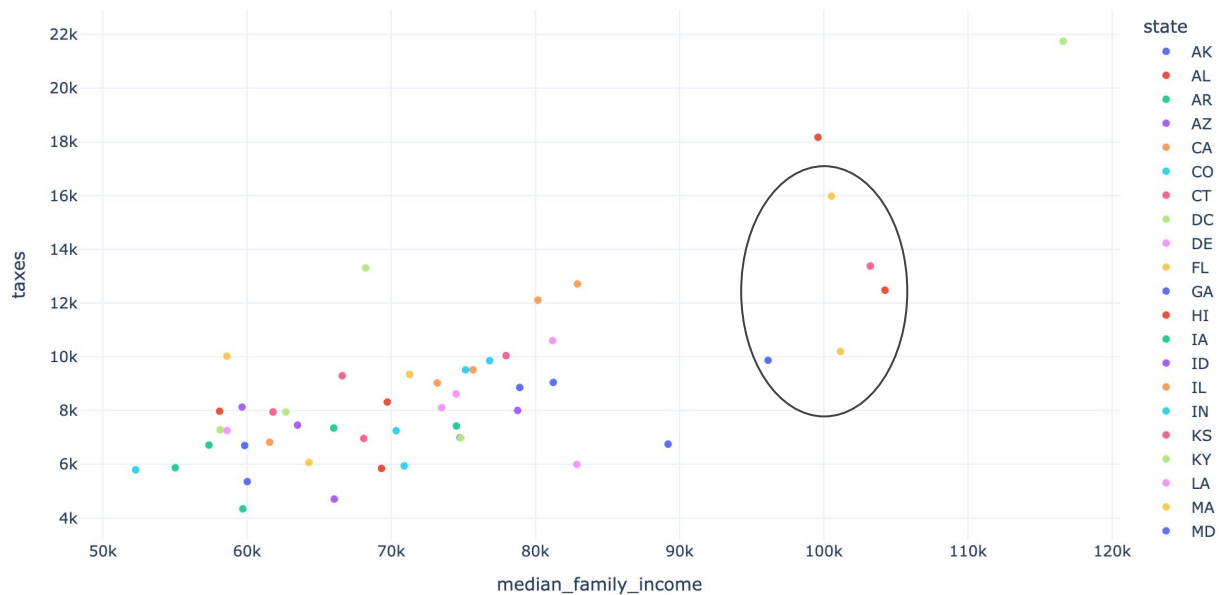
Transportation Cost vs. Total Cost by State





Scatter Plot - Taxes vs. Median Family Income

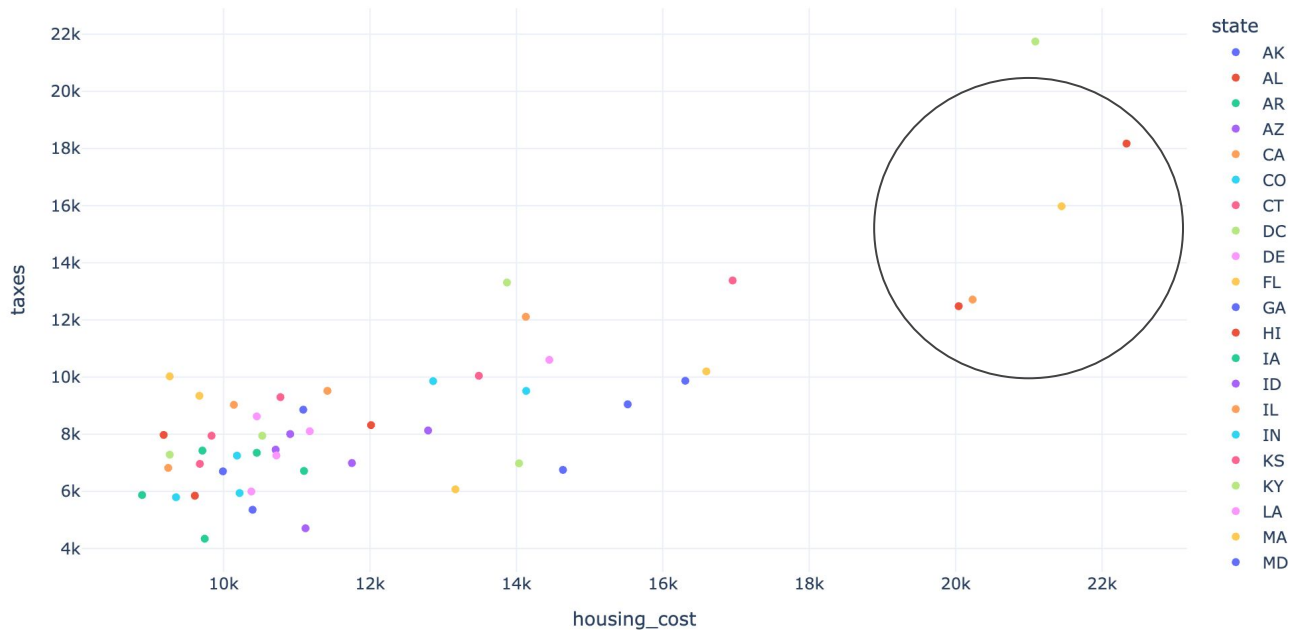
Taxes vs. Median Family Income by State





Scatter Plot - Taxes vs. Housing Cost

Taxes vs. Housing Cost by State





Scatter Plots Summary

- DC is the highest in at least one variable in every plot covered
- Taxes are strongly tied to most variables
- Geographical trends are apparent (Northeast US - High Income, Coastal States - High Housing Cost)



Most Expensive Counties - Childcare Cost

	state	county	childcare_cost
316	DC	District of Columbia	31940.49240
2011	NY	New York County	29620.24680
2021	NY	Queens County	28050.33432
2010	NY	Nassau County	27379.17840
2024	NY	Rockland County	27260.37720
2032	NY	Suffolk County	27122.55480
2020	NY	Putnam County	26750.40888
2040	NY	Westchester County	26661.03072
2004	NY	Kings County	26480.42088
2023	NY	Richmond County	25410.48264
1983	NY	Bronx County	24340.54296
1187	MA	Norfolk County	23606.17368

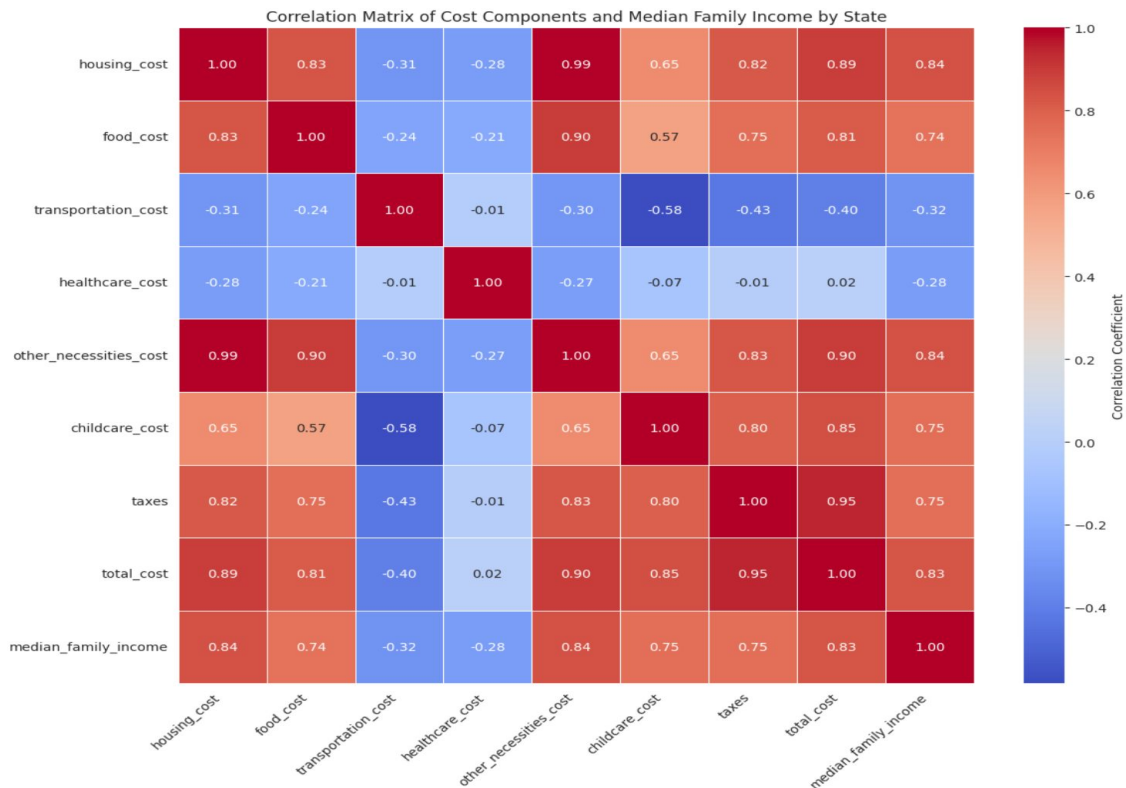


Most Expensive Counties - Taxes

	state	county	taxes
226	CA	San Mateo County	34738.964880
206	CA	Marin County	31637.529360
223	CA	San Francisco County	27174.135960
2011	NY	New York County	26517.054360
229	CA	Santa Cruz County	25831.416000
228	CA	Santa Clara County	24651.857760
2020	NY	Putnam County	24352.030080
2024	NY	Rockland County	23463.096600
2010	NY	Nassau County	23351.618064
2032	NY	Suffolk County	23094.638268
2021	NY	Queens County	22807.598208
1187	MA	Norfolk County	22347.509448

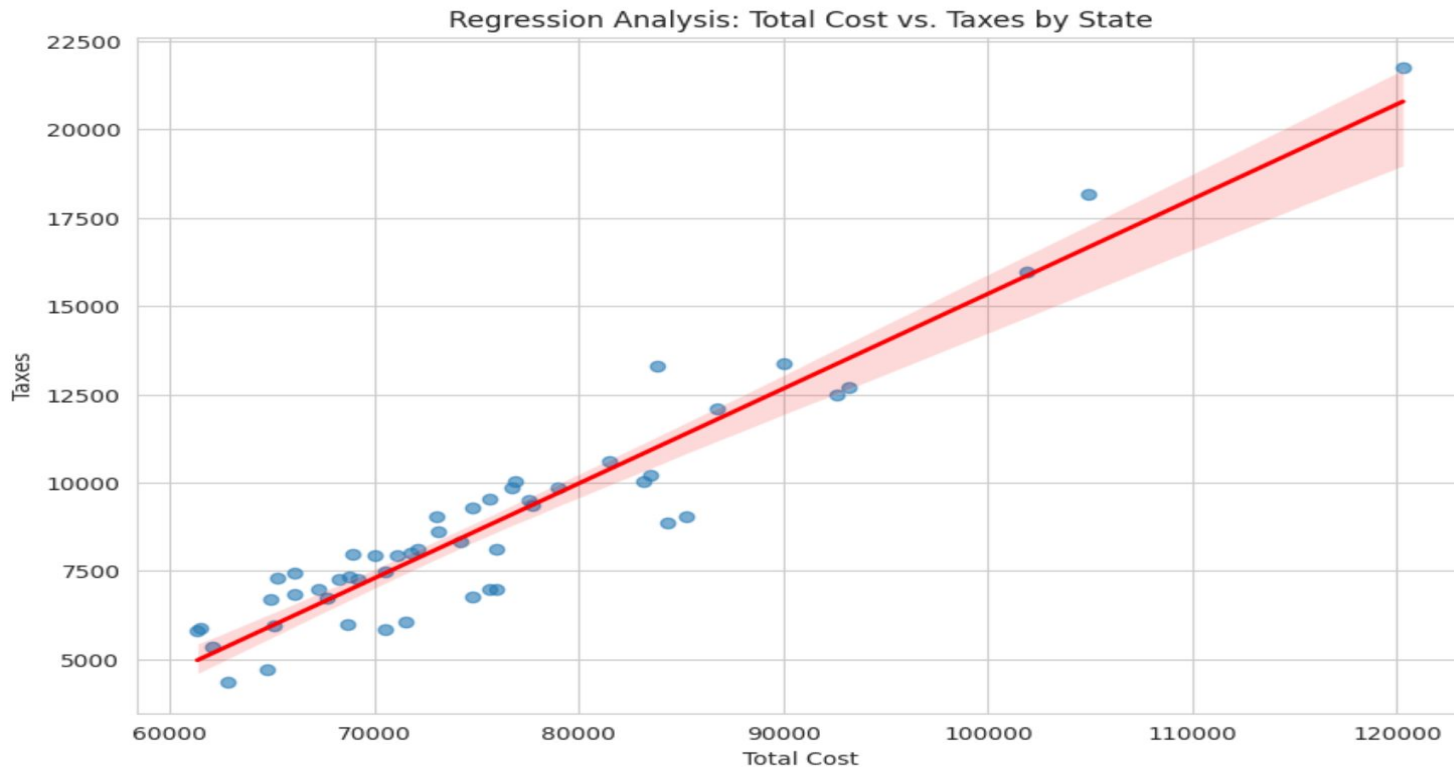


Correlation Matrix





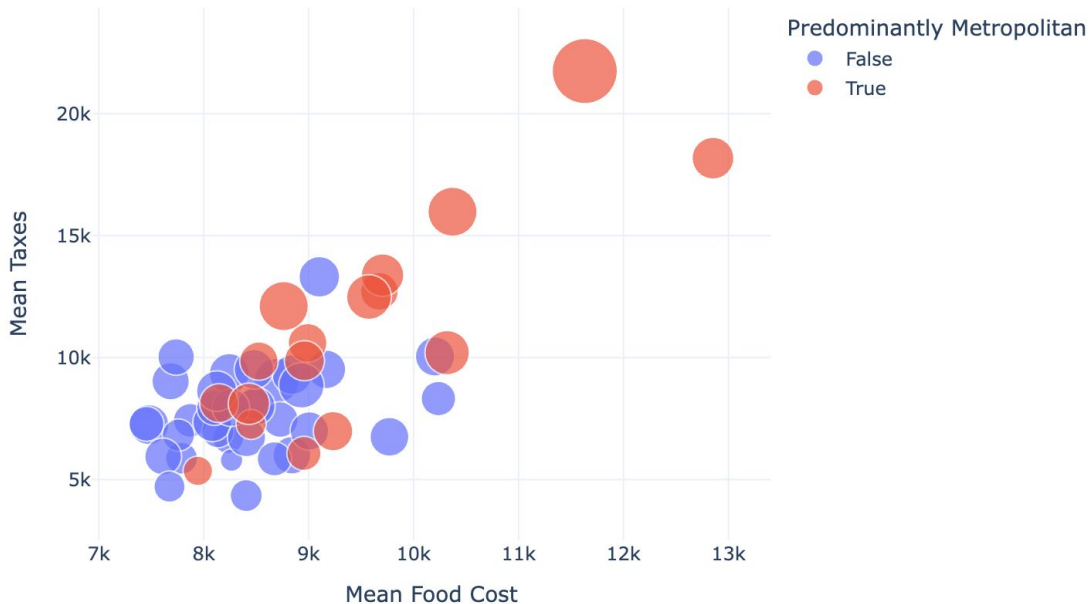
Regression Line - Taxes vs. Total Cost





Bubble Plot - Food Cost vs. Taxes (Size: Childcare Cost, Color: isMetro)

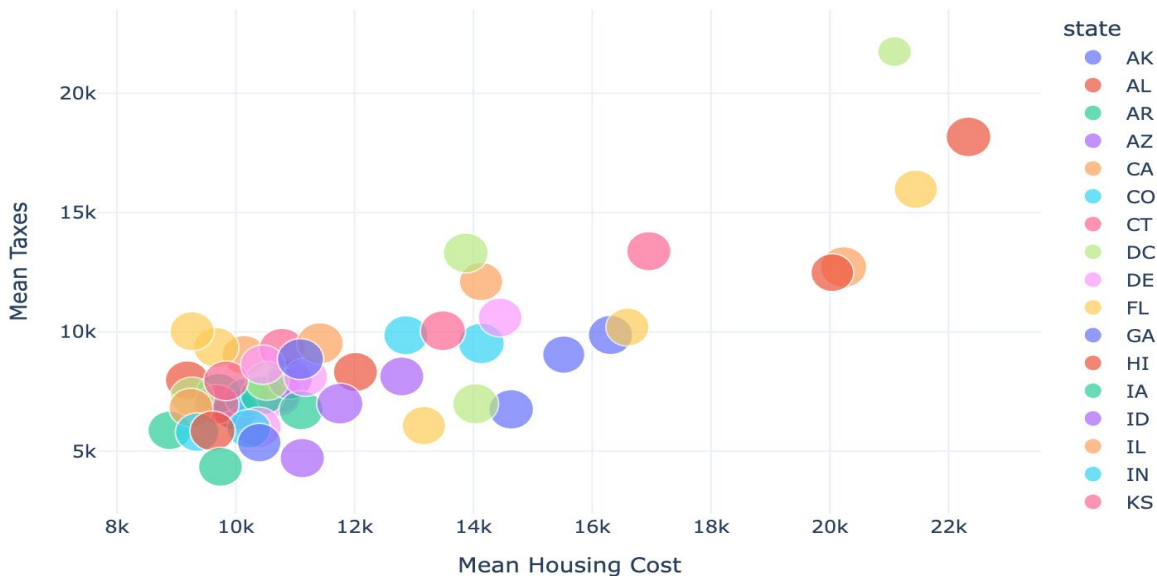
Food Cost vs. Taxes by State (Size: Childcare Cost, Color: isMetro)





Bubble Plot - Housing Cost vs. Taxes (Size: Transportation Cost)

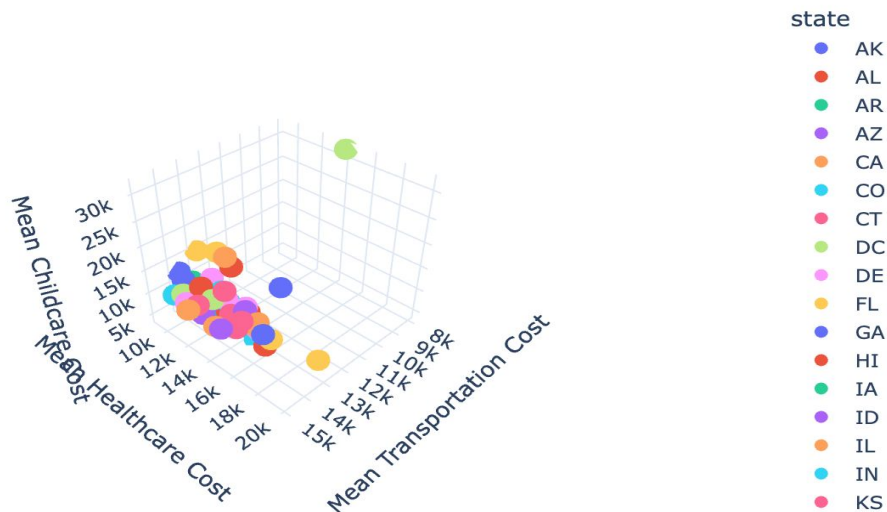
Housing Cost vs. Taxes by State (Size: Transportation Cost, Color: State)





3D Plot - Transportation, Healthcare, and Childcare Costs

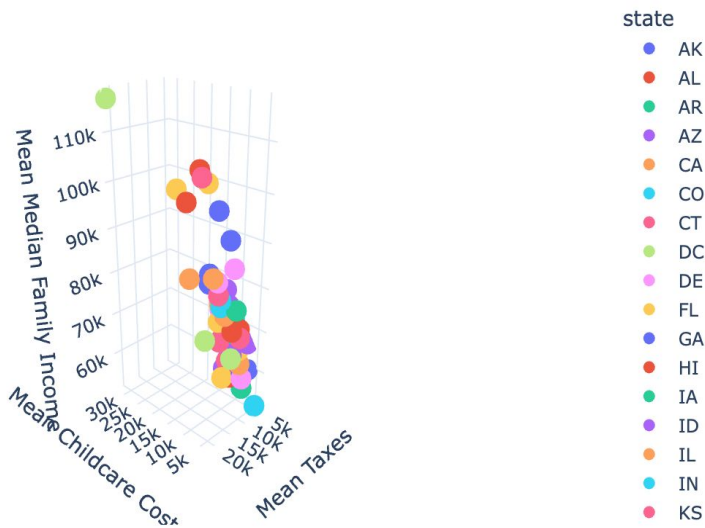
3D Plot of Transportation, Healthcare, and Childcare Costs by State





3D Plot - Childcare Cost, Taxes, and Median Family Income

3D Plot of Childcare Cost, Taxes, and Median Family Income by State



- Lots of variability in median family income
- Not much variability in other two variables
- DC is clear outlier with higher median family income, childcare cost, and taxes than anyone else



Advanced Visualizations Summary

- Taxes and median family income are closely correlated to many variables
- DC continues to be an outlier in most visualizations
- Clear concentrated groupings in most visualizations as well



New Findings / Key Insights

- Taxes drive many other costs in the US
- Geographic clusters exist within this dataset
- There are clear outliers on most visualizations (DC, HI, CA, NY)
- Some variables have much higher variability than others in this dataset



Conclusion

- The Cost of Living (US) dataset showed extremely interesting trends/patterns
- Most variables in the dataset are extremely closely correlated to each other
- There are clear regional trends throughout the dataset
- Outliers are prevalent in this dataset
- The cost of living in the US is very complex and multidimensional
- Each visualization showed new insights